Performance Testing

Date	3 October 2026
Team ID	SWTID-2026-9282
Project Name	ComicCraft – AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	ComicCraft Story Generation API – `POST /generate`
Test Environment	Local
Test Date	02-10-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Multiple users submit story prompts to the **Story Generation API (`POST /generate`)**.	5	60	The API should process the requests successfully and generate comic story content without application failure.
2	Multiple users request **five-panel comic generation (`POST /generate-comic/json`)**.	3	120	The system should generate the requested comic panels and return the response successfully.
3	Multiple users access the **ComicCraft homepage and API documentation (`/docs`)** simultaneously.	10	60	The application pages and API documentation should remain accessible without errors.
4	Multiple users perform comic generation requests continuously during the test period.	5	120	The application should continue processing requests without crashing or becoming unavailable.

Step 3: Performance Test Results

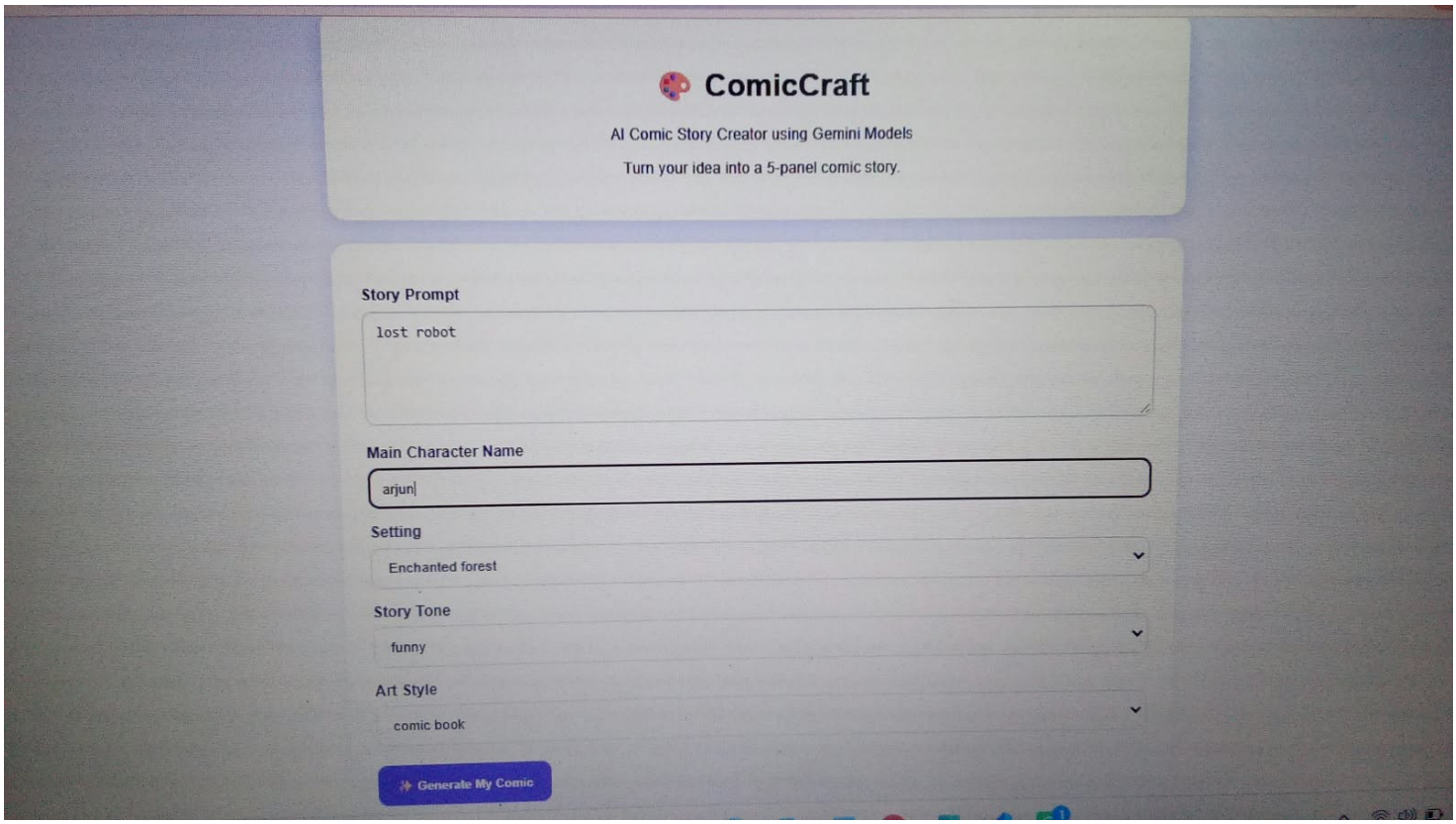
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	Not measured	Not tested	Not measured
2	Response Time (Max)	< 5 seconds	Not measured	Not tested	Not measured
3	Throughput (Req/sec)	To be measured	Not measured	Not tested	Not measured
4	Error Rate	< 1%	Not measured	Not tested	Not measured
5	CPU Utilization	< 80%	Not measured	Not tested	Not monitored
6	Memory Utilization	< 80%	Not measured	Not tested	Not monitored

Step 4: Observations & Analysis

S.NO	OBSERVATION / ANALYSIS	RESULT
1	Story generation was tested with different prompts.	Story generated successfully
2	Five-panel comic generation was tested.	Comic panels generated successfully
3	AI image generation was tested for comic panels.	Images generated successfully
4	PDF export and download were tested.	PDF generated successfully
5	FastAPI `/docs` was tested for API access.	API documentation accessible
6	Application stability was observed during testing.	Application worked without crash

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):


ComicCraft
AI Comic Story Creator using Gemini Models
Turn your idea into a 5-panel comic story.
Story Prompt
lost robot
Main Character Name
arjun
Setting
Enchanted forest
Story Tone
funny
Art Style
comic book
Generate My Comic